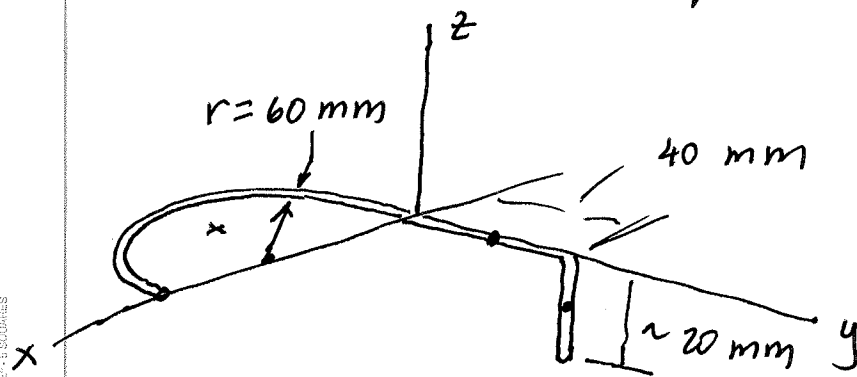
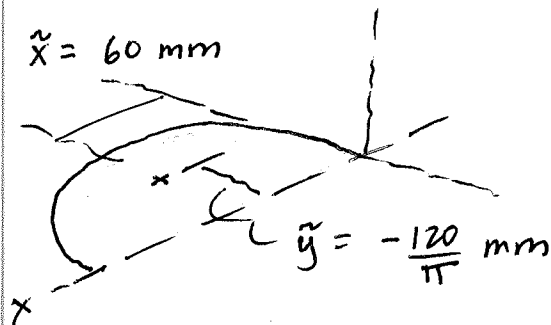


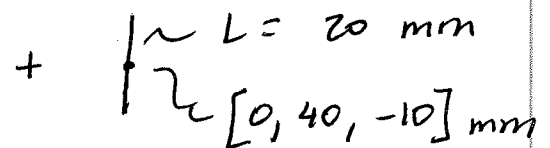
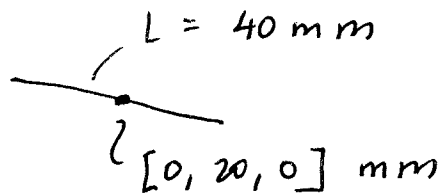


Find the
 $(\bar{x}, \bar{y}, \bar{z})$
centroid
of this
wire.



$$L = \frac{2\pi R}{2} = \pi R = 60\pi \text{ mm}$$

$$\text{Centroid: } = \frac{2R}{\pi} = \frac{2(60)}{\pi} = \frac{120}{\pi}$$



$$L_T = 60\pi + 40 + 20 = \boxed{248.5 \text{ mm} = L_T}$$

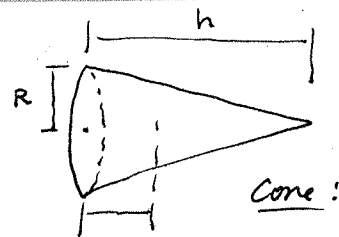
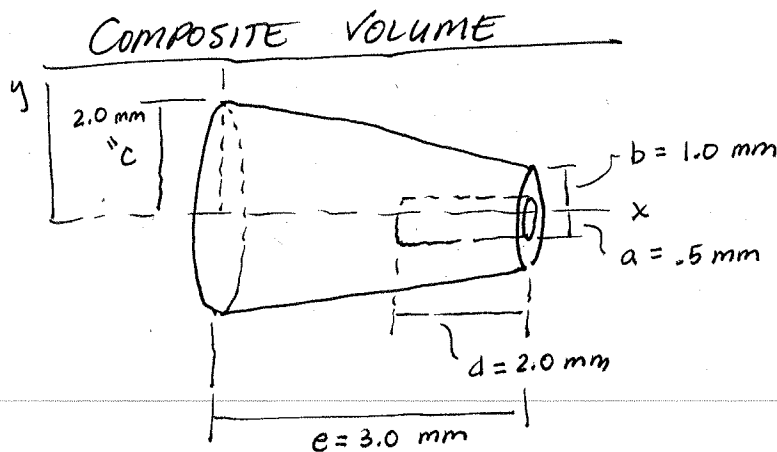
$$\bar{x} = \frac{\sum \tilde{x}_i L_i}{L_T} = \frac{60(60\pi) + 0 + 0}{248.5} = \boxed{45.5 \text{ mm} = \bar{x}}$$

$$\bar{y} = \frac{\sum y_i L_i}{L_T} = \frac{\left(-\frac{120}{\pi}\right)(60\pi) + 20(40) + 40(20)}{248.5}$$

$$\boxed{\bar{y} = -22.54 \text{ mm}}$$

$$\bar{z} = \frac{\sum z_i L_i}{L_T} = \frac{0 + 0 + (-10)(20)}{248.5}$$

$$\boxed{\bar{z} = -.8048 \text{ mm}}$$



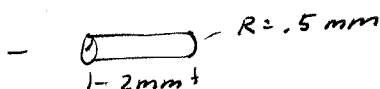
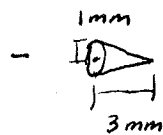
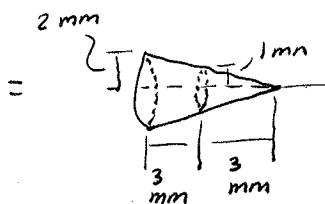
$$\bar{x} = h/4$$

$$V = \frac{1}{3}\pi R^2 h$$



Cylinder: $\bar{x} = h/2$

$$V_0 = \pi R^2 \cdot h$$



$$\bar{x} = \frac{\sum \tilde{x}_i V_i}{\sum V_i} = \frac{\left(\frac{6}{4}\right)(24\pi) + \left(3 + \frac{3}{4}\right)(-\pi) + (3-1)(-.5\pi)}{\underbrace{\frac{1}{3}\pi(2^2)(6)}_{24\pi} - \underbrace{\frac{1}{3}\pi(1)^2(3)}_{-\pi} - \underbrace{\pi(.5)^2(2)}_{-.5\pi}}$$

$$\bar{x} = \frac{98.175 \text{ mm}^4}{70.686 \text{ mm}^3}$$

$$\bar{x} = 1.389 \text{ mm}$$

$$\bar{y} = 0 \text{ because } x \text{ axis is an axis of symmetry.}$$

Notice I did not build a table on this problem. The first time you work a composite centroid problem it is best to build a table. Once you get comfortable w/ the procedure, a table can be too slow, and it is really not necessary.